

Werner Bisschoff

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Embedded Systems & Real-Time Software Engineer

Professional Summary

Rigorous Systems Engineer with 5+ years of experience engineering real-time, event-driven software architectures (C/C++, FreeRTOS) and secure, deterministic communication protocols. Specializes in designing robust firmware, high-performance IPC mechanisms, and hardware-in-the-loop simulation frameworks. Governed by strict Spec-Driven Development (SDD) methodologies and automated validation pipelines to eliminate runtime faults in production environments.

Experience

Principal Systems Architect , Cape Town Divergent Tabletop | Jul 2025

- Integrated CRDT synchronization layer (y_ex) for real-time modifications; actor-based buffering with sub-50ms sync latency.
- Designed dynamic BEAM supervision tree with :one_for_one strategy ensuring 99.9% fault isolation.
- Implemented threshold-based buffer flush and heap hibernation (hibernate_after), reducing memory footprint by 60%.

Embedded Systems & Integration Engineer , Cape Town FARO Africa | Aug 2024 – Nov 2025

- Designed ISO 14443-4 NFC communication architectures for e-paper display nodes and card reader matrices.
- Built Python-based hardware peripheral simulators to catch edge-case device state corruptions prior to flashing.
- Introduced LLM-assisted development workflows to improve debugging speed and code review throughput.

Firmware Engineer (Contract) , Remote Ingenics Digital GmbH (through ViVa Outsourcing) | Mar 2023 – May 2024

- Architected deterministic FSMs on dual-core ESP32 using FreeRTOS for concurrent sensor arrays.
- Developed Cap'n Proto IPC over serial lines, achieving sub-millisecond serialization latency.

Software Developer , Cape Town UMAN Technologies | Mar 2021 – Dec 2022

- Implemented RPC services using SOME/IP protocol; reduced bottlenecks with perf.
- Built IPC/RPC layers in C++ and Python using Cap'n Proto; designed node trees for IPC interface.

Junior Lecturer , Potchefstroom North-West University | Feb 2020 – Dec 2020

- Lectured Python and C++ programming for first-year IT students.

Technical Application Scope

Primary	C++ FreeRTOS ESP32/ESP-IDF NFC APDU (ISO 14443-4)
Secondary	NimBLE UART/SPI/I2C TinyFrame Hardware-in-the-Loop (HIL) Testing
Foundational	Assembly Digital Logic System Modeling
Other	ERPNext (Integration-focused) Expo Mobile

Education

B.Eng. Computer and Electronic Engineering North-West University | 2020

Focus on embedded systems, software engineering, and electronic design · Developed NFC payment protocol emulator (Kotlin/Android), STM32 firmware (C), PID motor controller (Arduino)